

Integration of Hybrid Thin-Film Lithium Niobate Electro-optic Modulators on a Wafer-scale Silicon Nitride Photonics Platform

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Abstract—This work achieves the desirable combination of low-pressure chemical vapor deposition silicon nitride (SiN) multi-layer photonic integrated circuits (PICs) with high-performance thin-film lithium niobate electro-optic (EO) modulators, made in a scalable foundry fabrication process using conventional low-resistivity silicon wafers. Operational parameters such as bandwidth, voltage, insertion loss and extinction ratio are reported for representative EO Mach-Zehnder modulator devices. The EO devices are seamlessly connected to low-loss SiN waveguides which have been used for superior performance in a wide range of PIC applications.

Index Terms—Integrated optics, Electro-optic modulation, Lithium Niobate, Silicon Nitride, Semiconductor fabrication.

I. INTRODUCTION

Lithium Niobate (LN) has many favorable electro-optic (EO) properties, such as a high Curie temperature, batch-to-batch uniformity, well-understood domain properties, and support for a wide range of optical wavelengths and radio frequencies (RF) [1]. Recent interest in lithium niobate on insulator (LNOI) and thin-film LN (TFLN) photonics [2], [3] has led to high-performance modulators using a variety of designs and fabrication approaches [4]–[9]. However, LN etching cannot be done in most silicon (Si) foundries because of potential contamination and the need for special processing [10]. In contrast, Si photonic EO devices, which are made in Si foundries, have shown limited performance with lower modulation amplitude at higher speeds, and limited linearity and optical power handling due to free-carrier absorption and two-photon absorption [11]–[13]. Therefore, an important challenge now is to make high-performance integrated TFLN EO devices (based on the Pockels effect) leveraging as much as possible industry-standard Si wafers and processes, which could lead to greater scalability and benefit wider segments of the photonics industry [14]. Here, we report the wafer-scale heterogeneous integration of high-performance hybrid TFLN modulators with multi-layer silicon nitride (SiN) photonic integrated circuits (PICs) on conventional Si wafers (see Fig. 1).

This work uses the hybrid-mode concept, in which the light is distributed among two (or more) waveguide core materials, one of which is a thin-film slab of LN, and the other(s) can be dielectrics or semiconductors (e.g., SiN or Si) which are patterned into strip waveguides [4], [7], [15],

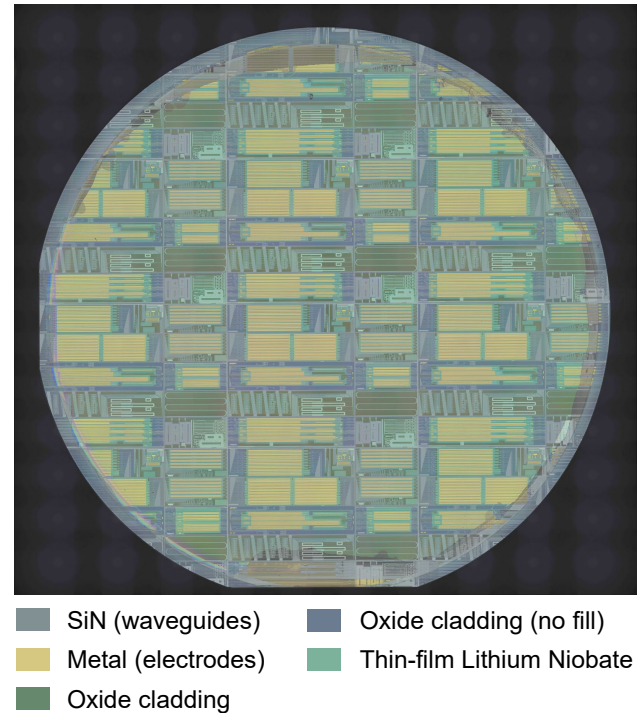


Fig. 1. A photograph of the fabricated wafer (100 mm diameter) with thin-film Lithium Niobate (TFLN) direct-bonded to multi-layer silicon nitride (SiN) photonic integrated circuits (PICs).

[16]. This design can be contrasted [17], [18] to one in which the LN slab was loaded by a dielectric material to form a rib waveguide [19]–[22]. In the hybrid-mode devices such as those discussed here, the waveguide core materials are separated by a thin cladding layer, such as silicon dioxide, and the strip waveguides in the dielectric material are used for passive photonic components without LN, facilitating seamless integration with a larger PIC. The hybrid-mode design can be implemented using bonding and does not require precise alignment during film transfer [23], [24], in contrast to micro-transfer printing [25], [26], and does not require a smooth and damage-free LN waveguide etching or milling process, in contrast to devices developed on an LNOI platform [6], [9], [27]–[29]. Furthermore, heterogeneous integration of thin-film EO crystals is a powerful tool for building upon existing PIC technology without having to start from scratch, since many components of a mature process design kit (PDK) in Si or SiN can be reused.

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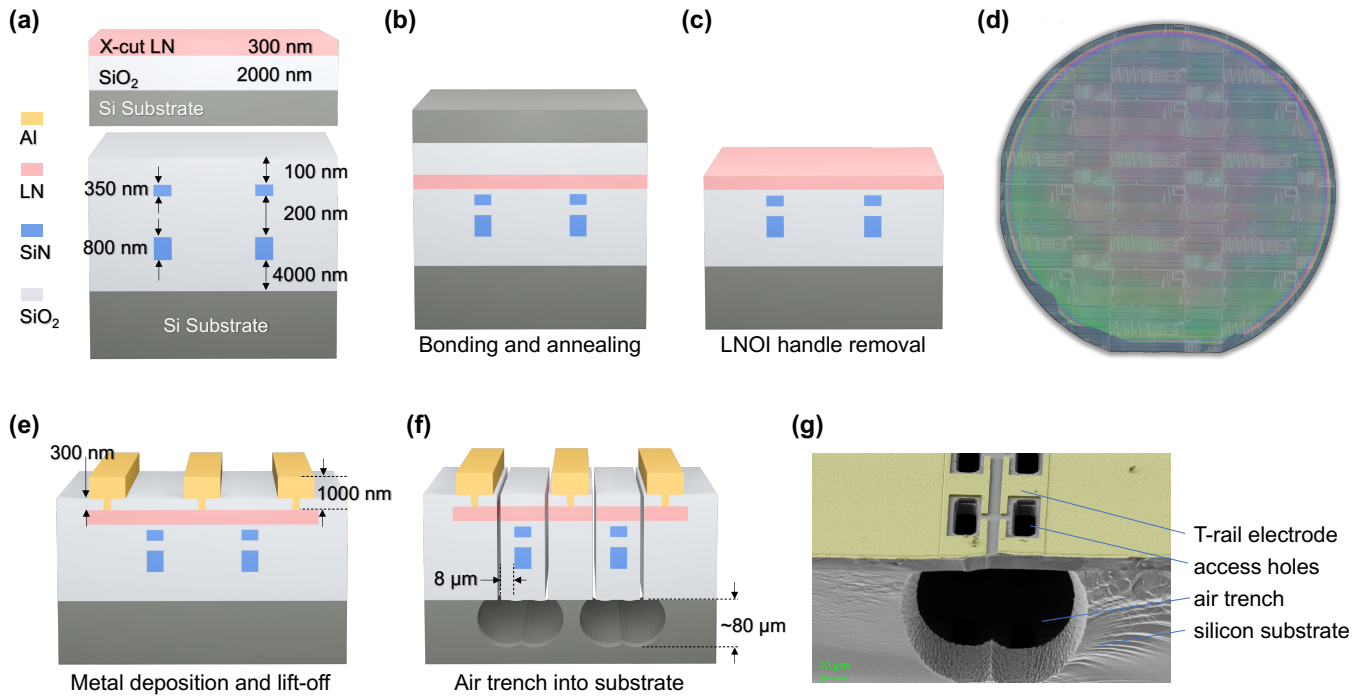


Fig. 2. Process flow diagram showing the main steps in panels (a)–(f) in the fabrication process. (Schematic drawings, not to scale.) (d) shows the bonded wafer pair after removal of the Si substrate of the LN wafer without significant bonding defects. (g) Cross-sectional scanning-electron microscope (SEM) image showing the trenches etched into the Si substrate.

II. FABRICATION

Thick layers of high-quality SiN, though desirable for superior device performance, cannot be deposited directly on LNOI wafers. A low-pressure chemical vapor deposition (LPCVD) SiN process with a low hydrogen content in the film typically results in the best optical properties at wavelengths around 1.5 μm [30] compared to plasma-enhanced chemical vapor deposition (PECVD) [16], [31]–[33] or sputtering of SiN [34], for example. However, optimized LPCVD SiN requires a temperature over 800 °C [35], which exceeds the maximum recommended for LNOI wafers before damage, about 550 °C [36]. Thus, only relatively thin SiN layers can be achieved by LPCVD deposition on LNOI [37]. This would hamper the performance of a larger PIC that requires dispersion-engineered low-loss waveguides and thick SiN. We resolve this incompatibility by optimizing the SiN process separately and then transferring the LN film by bonding the two wafers at a low temperature.

Figure 2 describes the fabrication process, which starts with a 200 mm diameter silicon (Si) handle wafer, with a thickness of about 700 μm and resistivity of about 5 Ω.cm measured using a four-point probe method. The two SiN layers, which we will label SiN1 and SiN2 in this discussion, were fabricated using LIGENTEC's proprietary technology with high-quality silicon nitride deposited using an LPCVD process (for additional processing details, readers may contact LIGENTEC). SiN1 has a thickness of 800 nm, and is used for various PIC components such as feeder waveguides, or directional couplers, and SiN2 has a thickness of 350 nm and is used primarily for the hybrid mode in the phase-shift

EO segment. A planar bondable surface was achieved using chemical-mechanical polishing (CMP). After processing, the wafer was cored down to a diameter of 100 mm to match the lithium-niobate-on-insulator (LNOI) wafers used here. Lithium-Niobate-On-Insulator (LNOI) wafers (100 mm diameter) with 300 nm thickness x-cut LN on 2 μm thick buried oxide, and an approximately 500 μm thick Si handle were commercially procured (NanoLN, Jinan Jingzheng Electronics Co., Ltd.).

After cleaning and surface activation, the Si and LNOI wafers were bonded using a hydrophilic direct-bond process [Fig. 2(b)]. The initial contact bonding was mediated by weak van der Waals forces between the exposed surfaces. After room temperature bonding, the bonded sample was annealed using temperature cycles up to 300 °C under applied pressure. The annealing process converts the hydrogen bonds into covalent bonds at the interface, thereby improving the bond strength [38]. No adhesives or polymers were used as the bonding layer. Adequate optical confinement can be achieved without etching of LN near the waveguide mode, and thus, the film quality is preserved. The handle and oxide layer of the LNOI wafer were removed using a combination of dry and wet etching steps [Fig. 2(c)]. About 95% of the SiN wafer area was bonded without significant bonding defects, apart from some minor delamination near the wafer flat, where there are no useful designs anyway [Fig. 2(d)]. The oxide layer above LN helps achieve an improved modulator design and fabrication process. The Aluminum (Al) electrodes were formed on top of the LN layer using deposition, lithography, and a dry-etch process [Fig. 2(e)].

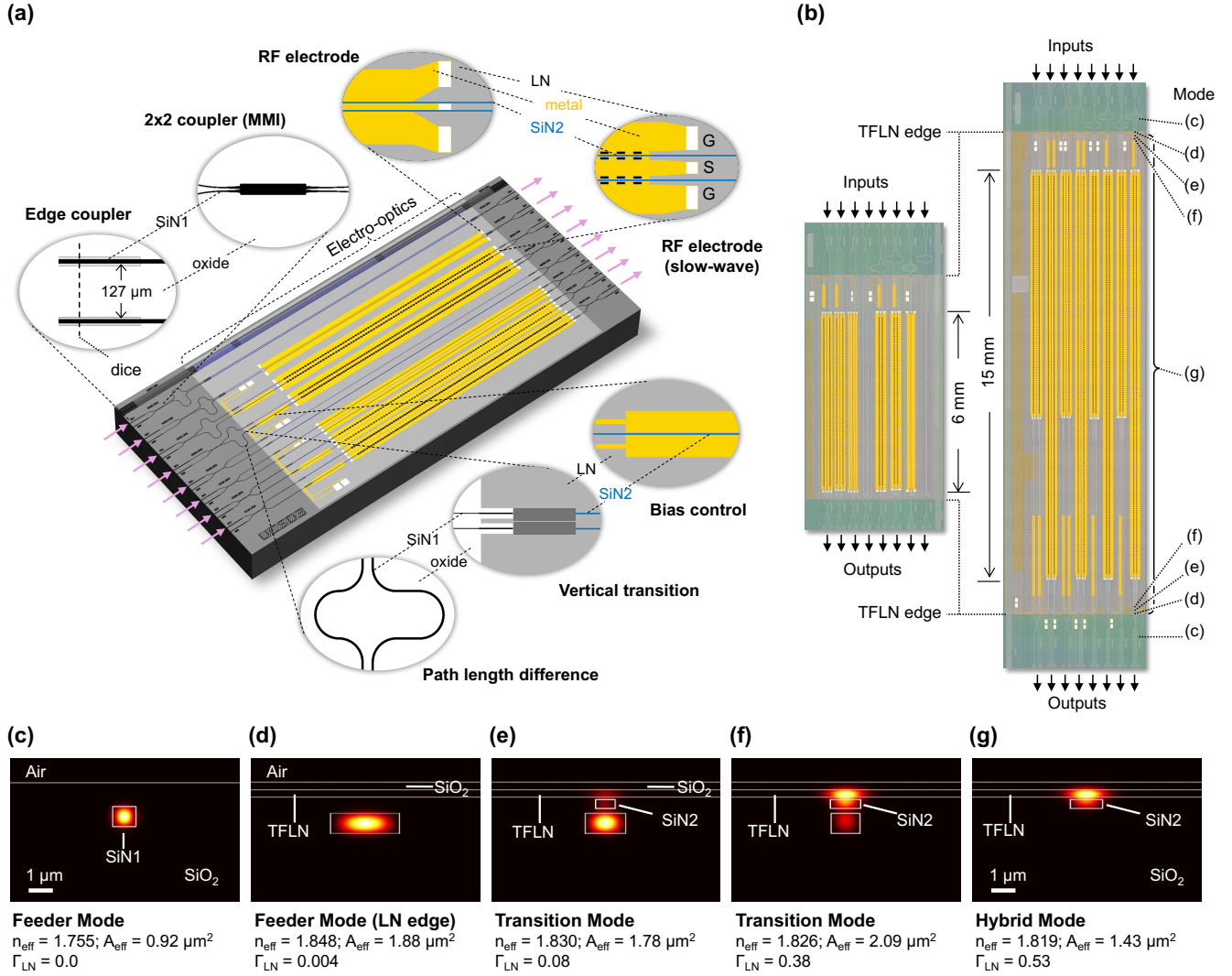


Fig. 3. (a) Diagram of the hybrid microchips showing various optical and RF building blocks. (b) Photograph of two microchips with multiple EO modulator devices. The bonded TFLN region is visible as a slightly lighter colored rectangle covering the central portion. (c)–(g), Simulations of the optical waveguide mode (cross-section) in the feeder, transition, and hybrid sections, which are indicated in panel (b). The optical effective index (n_{eff}), effective area (A_{eff}) and mode confinement fraction in LN (Γ_{LN}) are tabulated. Abbreviations: MMI: multi-mode interference, RF: radio frequency, SiN: silicon nitride, TFLN: thin-film lithium niobate.

Whereas much of TFLN EOM development uses specialized handle wafers such as high-resistivity Si or quartz, the SiN process utilized here uses standard low-resistivity Si wafers (about $5 \Omega\cdot\text{cm}$ after processing) which are much more common in industry. However, such wafers have a high RF loss and would limit the EOM bandwidth. To reduce the RF loss, we incorporated air trenches etched under the phase-shift sections where the RF and optical waves interact. Trenches have been used in various integrated photonics platforms e.g., to improve optical confinement [39], [40] and improve EO properties [29], [41]–[43]. Vertical trenches were formed from the LN surface through the oxide into the Si substrate as access holes for a dry-etch process which formed an undercut [Fig. 2(f)]. The trenches have lateral dimensions of about $30 \mu\text{m} \times 8 \mu\text{m}$ (length $\times$ width) and were about $5 \mu\text{m}$ in height. Inspection showed an undercut depth of about $80 \mu\text{m}$. They were placed at a large horizontal distance (about $8 \mu\text{m}$) from the SiN2

strip waveguide edge and do not affect the optical waveguide mode properties. These trenches were used to access the Si substrate for an isotropic dry etch process which locally undercut the phase-shift segments and also removed the LN over the passive sections where it is not needed. Because this substrate partial etch is over a limited area, and a substantial portion of the handle remains, we have not seen problems with mechanical robustness or in dicing. Figure 2(g) is a cross-sectional scanning-electron microscope (SEM) image showing the trenches etched into the Si substrate to lower the propagation loss of the traveling-wave microwave mode, as discussed later in this report. The completed wafer is shown in Fig. 1.

III. DESIGN

Here we describe the design of EO modulators in the push-pull Mach-Zehnder interferometer configuration using

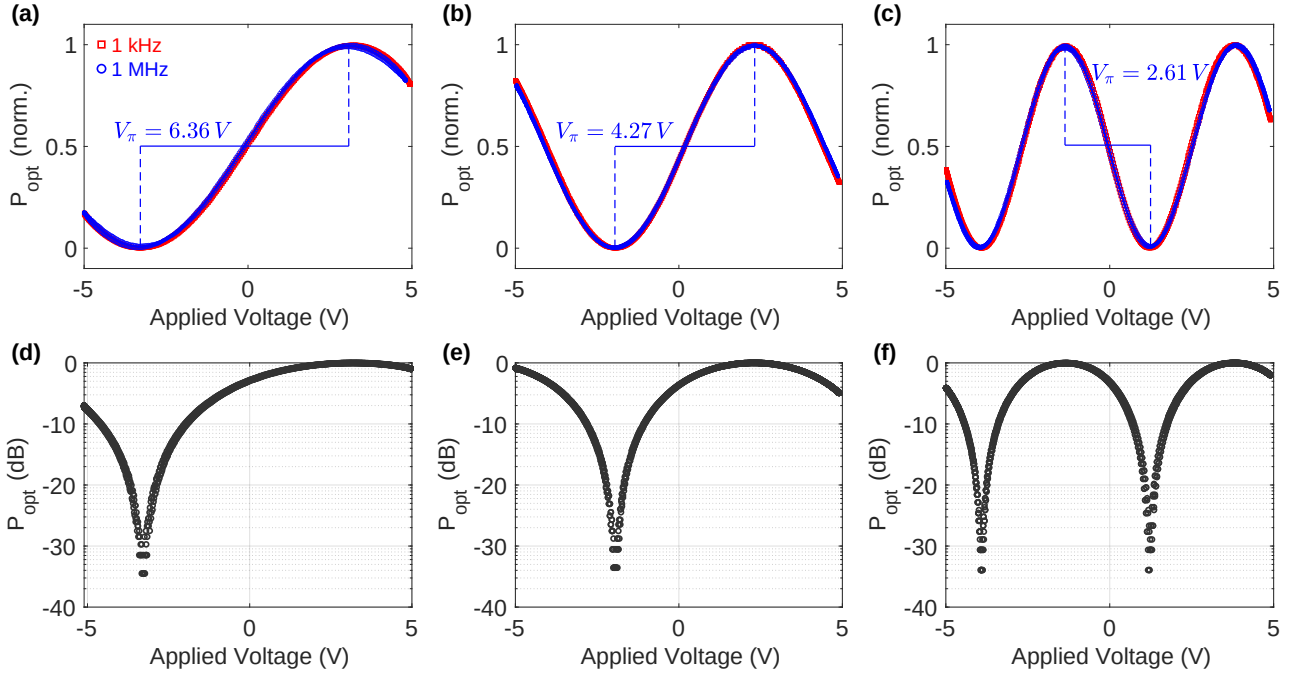


Fig. 4. (a, b, c) The normalized optical power measured as a function of the applied voltage at frequencies of 1 kHz (red trace) and 1 MHz (blue trace), for $L = 0.6$ cm in panel (a), 0.9 cm in panel (b) and 1.5 cm in panel (c). The measured half-wave voltage (V_π) is indicated in each case. (d, e, f) The normalized optical power (at a frequency of 1 kHz) is shown with a logarithmic vertical scale to quantify the extinction ratio, for $L = 0.6$ cm in panel (d), 0.9 cm in panel (e) and 1.5 cm in panel (f).

this platform. Figure 3(a) shows some of the main optical and RF building blocks of the hybrid PIC. Figures 3(b) show two dies singulated from the wafer with multiple such devices with different phase-shifter lengths $L = 0.6$ cm, 0.9 cm and 1.5 cm. The bonded LN film is visible as a slightly lighter rectangle on top of the SiN waveguides in the middle portion of the PIC.

This standard device serves as a reference and has been made using many different approaches, including etched LNOI waveguides, rib-loaded LNOI waveguides, and hybrid-mode TFLN waveguides (for review articles, see [44], [45]). In the hybrid-mode design, optical mode control is achieved by changing the SiN strip waveguide width, without etching LN [15]. The optical and microwave structures reported here were designed using software (Ansys Lumerical; and PathWave EM Design, Keysight Technologies) guided by insights from earlier work [46]. SiN waveguides which have the mode structure shown in Fig. 3(c) confine most of the light in the passive material, and were used in the feeder section and for waveguide bends, splitters and combiners. Waveguides which push light mostly into the LN region were used in the phase-shift section of the EOM device, using the mode structure shown in Fig. 3(g). The feeder waveguides were defined in the SiN1 layer, whereas the SiN strip waveguide that forms the hybrid mode with the TFLN slab was defined in the SiN2 layer. Compared to a single SiN layer design, the design with double SiN layers reduces the mode mismatch between the passive sections (no LN) and the EO sections (with LN), and results in better EO performance with lower

insertion loss [24]. The strip waveguides in the SiN1 layer have dimensions of $1 \mu\text{m} \times 0.8 \mu\text{m}$ (width $\times$ thickness), and those in the SiN2 layer have dimensions of $1.4 \mu\text{m} \times 0.35 \mu\text{m}$. The buffer oxide layer below SiN1 has a nominal thickness of $4 \mu\text{m}$, and the top oxide layer above SiN2 has a nominal thickness of $0.1 \mu\text{m}$. The SiN1 layer also contains several PIC components, such as the waveguide-to-fiber couplers at the edge of the chip, the 2×2 multi-mode interference (MMI) directional couplers (length about $100 \mu\text{m}$), and an optical path length difference segment in the passive region outside the LN (thus creating an asymmetric MZI structure). The feeder waveguides in SiN1 pass the abrupt LN boundary and continue a short distance into the LN-covered region. This is followed by an adiabatic taper between the SiN1 and SiN2 layers, which was formed by decreasing the width of the SiN1 waveguide in the lower layer while simultaneously increasing the width of the SiN2 waveguide in the upper layer [47]. Such tapers can be used for vertical interlayer mode transitions with low reflection and scattering [48]. The final width of the SiN2 waveguide is chosen to achieve index matching with the RF mode for optimal EO performance. As shown in Figs. 3(c)-(g), the optical mode shifts from the SiN1 layer to transitional modes and finally to the hybrid mode [Fig. 3(g)] with most of the light in LN, which is used in the EO phase-shift segments. The SiN1 waveguide has the smallest optical effective area ($A_{\text{eff}} = 0.92 \mu\text{m}^2$), whereas a representative mode in the transition section and the hybrid mode have an area of $1.78 \mu\text{m}^2$ and $1.43 \mu\text{m}^2$, respectively. The confinement factor Γ stated in Figs. 3(c)-(g) is defined by

the spatial distribution of the magnitude of the Poynting vector, $\mathbf{S} = \mathbf{E} \times \mathbf{H}^*$ where $\mathbf{E}$ and $\mathbf{H}$ are the electric and magnetic field distributions. The hybrid mode has $\Gamma_{\text{LN}} \approx 53\%$ of the optical power in LN.

The electrode pattern was designed to match the optical group index of the hybrid mode near 1550 nm wavelengths to the RF phase refractive index over a broad range of RF modulation frequencies up to 100 GHz (target value 2.11). One of the designs uses a signal-ground gap of $5 \mu\text{m}$ with a capacitively-loaded T-rail electrode design for RF-optical phase matching, similar to other work [46], [49]. Some devices include a section for electrical bias control (which was not necessary for the measurements reported here). A path length difference (PLD) segment was designed in the “passives” section before the EO section, as shown in Fig. 3(a). In an asymmetric MZI structure, the PLD causes a periodicity in the transmission spectrum. We used this information to determine the optimal wavelength at which to test the performance of the EOM, by selecting a wavelength at which the transmission is biased 3 dB below the peak transmission. This corresponds to the linear-most portion of the inherently-nonlinear modulation transfer function (MTF) of a MZI-based modulator device [50]. The devices reported here do not exhaust all the design choices that are available. By using the same SiN1 layer as in LIGENTEC’s existing platform, these devices can be seamlessly connected from the EO section to other PIC components outside the LN region.

IV. RESULTS

A. Optical Insertion Loss

Several dies with the same design were diced from the wafer and were tested using an optoelectronic probe station at room temperature. Continuous-wave light was coupled to the waveguides using a tunable semiconductor laser source (81640B; Keysight Technologies) and lensed tapered fibers with a spot size of about $3.5 \mu\text{m}$ (Oz Optics). The on-chip optical power was about -3 dBm (0.5 mW). The insertion loss (IL) of the EOM was measured by comparing the optical transmission through the EOM to the transmission through a passive waveguide formed in the SiN1 layer. Both devices use similar waveguide-fiber edge couplers, but the passive waveguide does not include transitions to and from the SiN2 layer, or the sub-components of the modulator, such as the 3-dB input and output couplers and the phase-shift segments. Since the modulator is based on a MZI which includes a path-length difference segment, the values at the peak of the spectra near 1550 nm were taken as representative of its transmission. For the $L = 0.6 \text{ cm}$ device, the measured average on-chip IL (and one standard deviation) for devices across four dies was 4 dB (0.8 dB). The observed variation in IL from one die to another is a consequence of the varying layer thicknesses and fidelity-to-design of devices fabricated on the edges of the wafer compared to those in the center.

On a chip containing modulators, we measured a set of four waveguides of different SiN1/SiN2 lengths (SiN1 lengths ranging from 3 mm to 2.1 cm, and SiN2 lengths ranging from 0 cm to 1.76 cm). By finding the minimum linear least-squares

error (LLSE) solution of the resulting set of equations, we inferred that the propagation loss of the SiN1 waveguide mode (passive) is 0.27 dB.cm^{-1} and of the SiN2 waveguide (hybrid mode) is 1.42 dB.cm^{-1} at around 1550 nm. The low loss of the SiN1 layer was confirmed by separate measurements as described below. The higher loss in the SiN2 layer can be attributed to damage of this layer during the etching of the LN layer, and material absorption in the top oxide layer. A more selective LN removal process to avoid such damage is under development, which should result in improved propagation losses. Also using these waveguides, the LLSE estimate of the transition loss from SiN1 to/from SiN2 was 0.20 dB, which is consistent with the range of values measured on separate test structures stated below.

On separate chips, waveguide-coupled microring resonators with a waveguide width of $1.7 \mu\text{m}$ were formed in the SiN1 layer with a free spectral range of 150 GHz and measured to have a loaded quality factor at critical coupling of about 1.2 million, from which we infer a propagation loss of about 0.2 dB.cm^{-1} at 1550 nm across the whole wafer, which is similar to the measured value described in the previous paragraph. No significant change in loss was observed by comparing the transmission of devices before and after LN bonding and processing.

Also on separate chips selected from different portions of the wafer, measurements of separate test structures consisting of different numbers of repeated transitions of SiN1 to/from SiN2 waveguide segments showed that the IL for the SiN1-SiN2 waveguide transition is low, and varies from 0.05 dB to 0.25 dB per transition at 1550 nm. This range of values is consistent with the value (0.2 dB) made on a set of four waveguides located in one portion of the wafer described above.

On a more recent fabrication run, the process was further optimized to reduce passive optical propagation losses in the hybrid mode. Through the characterization of delay lines in a cutback configuration, an average propagation loss $< 5 \text{ dB.m}^{-1}$ (decibels per meter) was measured around 1550 nm across a batch of 6 wafers. This highlights that the results obtained in this work can be further improved through iterative process optimization.

B. Half-Wave Voltage-Length Product ($V_\pi L$)

To measure the half-wave voltage (V_π), low-frequency signals generated by a pulse pattern generator (81110A, Agilent Technologies) were used to drive the modulator with a certain peak-to-peak voltage that exceeds the V_π value anticipated from design simulations. The modulated optical signal was detected with a linear photodetector (Model 1817-FC, Newport) and the electrical signal was recorded on an oscilloscope. This waveform was post-processed using software (Matlab) to map the optical transmission to the applied voltage, and to find V_π .

Fig. 4 shows the modulated signals, normalized to 0 dB peak transmission, as a function of applied voltage for the modulators. The graphs are shown for frequencies of 1 kHz and 1 MHz without a significant difference between the two

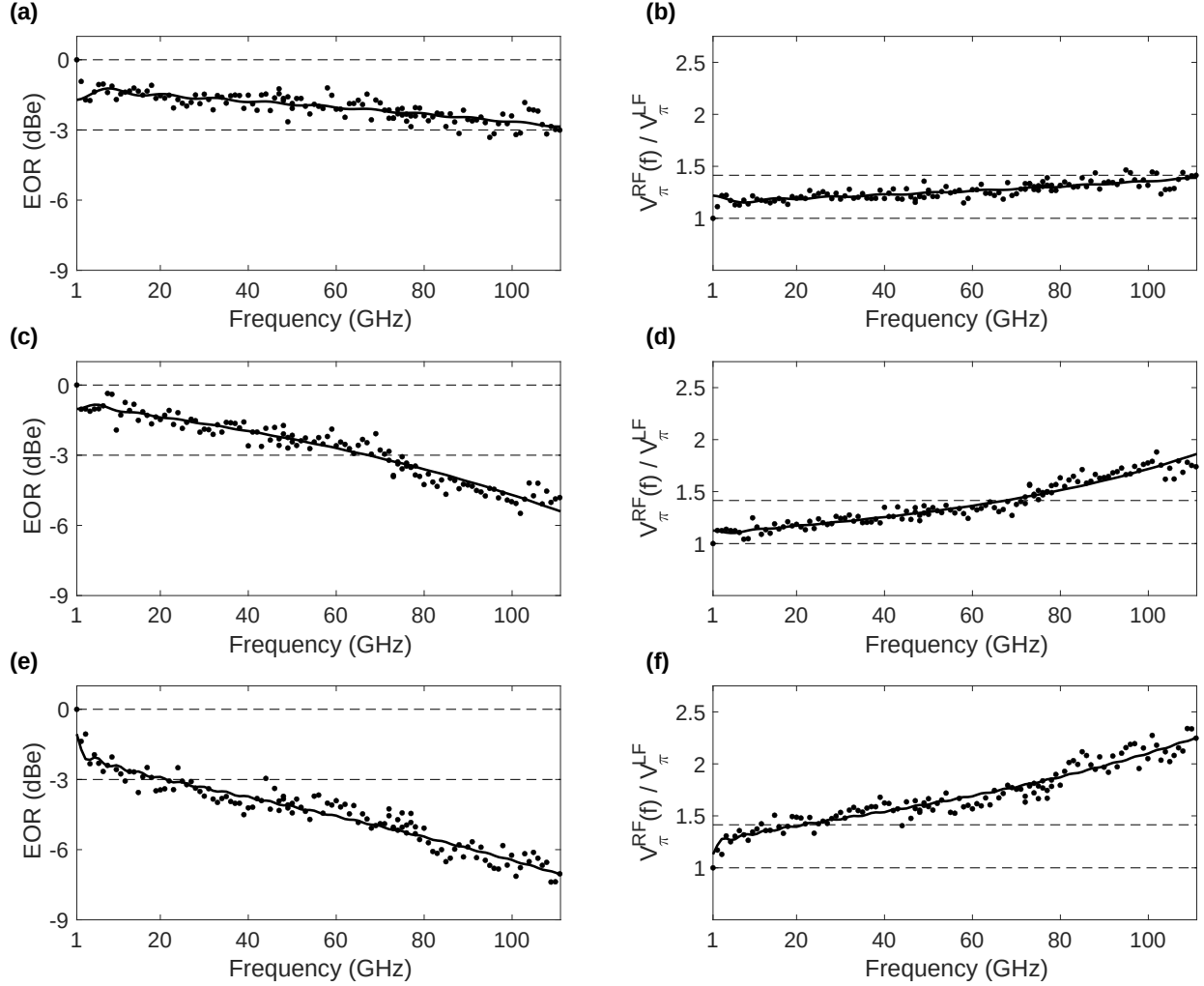


Fig. 5. (a, c, e) The EOR between 1 GHz and 111 GHz, normalized to the corresponding EOR value at 1 GHz, for modulators with $L = 0.6$ cm in panel (a), 0.9 cm in panel (c), and 1.5 cm in panel (e). The solid black line is a fit to the data using a theoretical model. (b, d, f), The normalized RF half-wave voltage as a function of driving frequency for modulators with $L = 0.6$ cm in panel (b), 0.9 cm in panel (d), and 1.5 cm in panel (f). The dashed lines show the 3-dB electro-optic modulation bandwidth, referenced to 1 GHz.

cases, or for other intermediate frequencies. The data was fit using a cosine-squared equation, resulting in a measured V_{π} of 6.4 V ($V_{\pi}L = 3.8$ V.cm) in both cases for the 0.6 cm modulator. These values were consistent across different chips selected from the wafer: the average (and one-standard deviation) $V_{\pi}L$ values measured across four chips were 3.83 V.cm (0.01 V.cm). For the longer modulator devices, the measured values on one chip were V_{π} of 4.3 V ($V_{\pi}L = 3.9$ V.cm) for the 0.9 cm modulator, and V_{π} of 2.6 V ($V_{\pi}L = 3.9$ V.cm) for the 1.5 cm modulator.

Plotting the 1 kHz data using a logarithmic scale for the vertical axis demonstrates an extinction ratio greater than 35 dB in all cases. The high ER shows that the input and output 3-dB couplers were well-balanced, and the losses in both arms of the push-pull MZI structure were also well-balanced after fabrication was completed.

The measured average $V_{\pi}L$ values of 3.8 V.cm can be

compared to the values expected from simulations using a standard value of $r_{33} = 30.8$ pm/V, which is 3.33 V.cm. Using a recently-reported value for TFLN at 1550 nm of $r_{33} = 26.9$ pm/V [51], the simulated value is 3.81 V.cm, which agrees with the measurement.

C. 3-dB EO Modulation Bandwidth

We used a modified modulation-sideband method for measuring the electro-optic response (EOR), and our previous work has discussed the similarity of this measurement method to others e.g., based on using a Lightwave Component Analyzer [46], [52]. These methods have given equivalent results over the frequency range (up to about 110 GHz) used here. Sinusoidal RF waveforms between 1 GHz and 50 GHz were generated using an RF sweeper (83651B, Hewlett Packard, Inc.). A high-resolution OSA (WaveAnalyzer 1500S, Finisar)

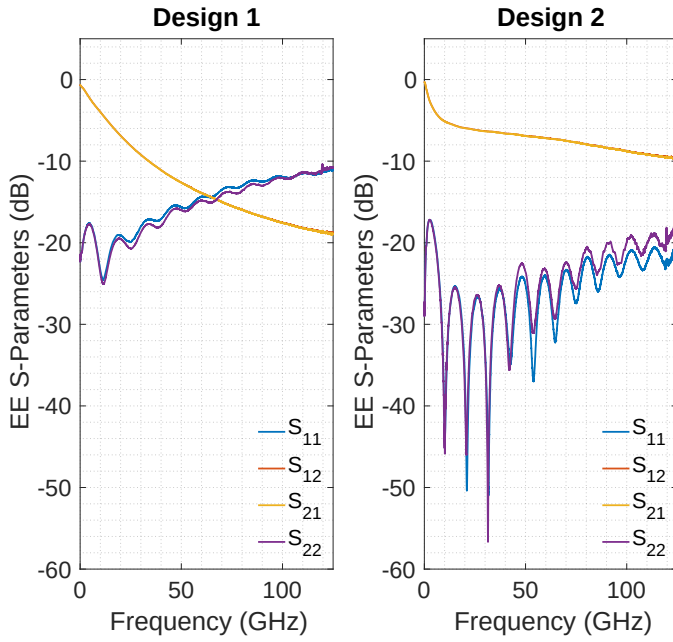


Fig. 6. The measured S-parameters (100 MHz to 125 GHz) of a Design 1 and a Design 2 EOM, as described in the text, on the same test chip. The $|S_{21}|^2$ and $|S_{12}|^2$ curves overlap.

was used to measure the spectrum of the modulated signals. High-speed ground-signal-ground RF probes rated to 110 GHz (Model i110, FormFactor) sourced and terminated the traveling-wave electrodes and were used with standard low-loss 1 mm RF cables and connectors. The EOR was determined by tracking the peaks of the carrier signal and the generated sidebands from modulation [53]–[56]. Active multipliers were used for the 47 GHz to 78 GHz band (SFA-503753420-15SF-E1, Eravant) and for the 72 GHz to 110 GHz band (SFA-753114616-10SF-E1, Eravant). The amount of RF power delivered to the chip was recorded using a calibrated thermal RF power meter (NRP110T.02, Rohde & Schwarz). The two frequency ranges over which the banded measurements overlap (47–50 GHz and 72–78 GHz) were used to stitch the EOR responses into the composite EOR response as shown in Fig. 5. This was accomplished by calculating a mean value for both traces in the overlapping frequency range, and vertically offsetting the second data set to the first curve. The solid line is a fit to the data using a transmission-line model of the hybrid TFLN EOM [46].

It is useful to plot the EOR as a frequency-dependent scaling of the half-wave voltage [57], given by $V_{\pi}(f) = V_{\pi}(\text{LF})10^{-m(f)/20}$, as a function of modulation frequency f , where $V_{\pi}(\text{LF})$ is the low-frequency half-wave voltage and $m(f)$ is the measured EOR. For a numerical estimate of $V_{\pi}(\text{LF})$, we use the half-wave voltage at 1 GHz and Fig. 5 shows the calculated RF $V_{\pi}(f)$ normalized to it. An increase of this ratio by a factor of $\sqrt{2}$ defines the modulation bandwidth because it correlates to the EOR of the modulator (and equivalently, the EO $|S_{21}|^2$ value) decreasing by 50%. This is shown by the dashed lines in Fig. 5. Thus, we can take as the 3-dB modulation bandwidths of these devices as 110 GHz ($L = 0.6$ cm), 70 GHz (0.9 cm) and 20 GHz (1.5 cm).

D. Electrical Characterization

The electrical response of some EOM devices was measured using a two-port Keysight PNA-X vector network analyzer (VNA) with broadband frequency extenders, allowing for RF signals from 100 MHz up to 125 GHz to be applied. Figure 6 shows the measured electrical S-parameters of two designs (Design 1 and Design 2), which are described in the next paragraph. The small artifact around 118 GHz visible in some of these traces was due to exceeding the nominal upper frequency of the GSG probes (110 GHz). We used different chips (from the same wafer) for the VNA and the EOR measurements and minor variations between Fig. 6 and Fig. 7 can be expected.

Both devices have the same phase-shift length, 0.6 cm. Design 1 uses coplanar waveguide (CPW) electrodes with a signal-ground gap distance of $G = 6 \mu\text{m}$. According to our simulations, Design 1 can achieve a good match between the RF phase and optical group indices of refraction up to high RF modulation bandwidths. However, as stated earlier, the devices were fabricated on a wafer with a low-resistivity Si substrate, for which the high-frequency RF propagation loss is too large to support high modulation bandwidths. Therefore, in Design 2, portions of the substrate were removed locally by etching air trenches as described earlier [see Fig. 2(g)] whereas no trenches were included with Design 1, in order to provide a comparison. We used periodically-patterned slow-wave electrodes (SWE) with $G = 5 \mu\text{m}$ to design for matching the RF optical index to the target optical group index (2.11).

Evidence of superior RF loss and better overall performance was seen in the measured S-parameters, which showed that the Design 2 device has a more gradual roll-off in $|S_{21}|^2$ than the Design 1 device. Also, the RF back-reflection ($|S_{11}|^2$) is significantly better (lower) for Design 2. To study the causes for this difference in performance, we calculated the RF index, loss, and impedance of the transmission lines from the measured S-parameters [58]. The results are shown in Fig. 7 for Design 1 using red lines, and for Design 2 using black lines. The Design 2 device is better in all three metrics: index matching, loss and impedance matching to 50Ω . Figure 7(a) shows that the dispersion of the RF index is lower for Design 2, and value of the index is closer to the target. However, even at 100 GHz, the Design 1 index is less than 10% smaller than the target, and is not the main difference for a low EOR bandwidth. Similarly, Fig. 7(c) shows that Design 2 has a better match to 50Ω , but Design 1 has a range of frequencies around 50 GHz where it is also a reasonable match. The important difference is that Design 2 has a lower RF propagation loss than Design 1 for RF frequencies above 10 GHz.

The dashed line in Fig. 7(b) is a fit to the RF loss using a skin-effect loss model which has a square-root dependence of loss on frequency. This model is not always a perfect fit for TFLN modulators throughout the three octaves of RF frequencies measured here, but it usually shows the trend at higher frequencies [46]. The fitted skin-effect loss coefficient of the Design 1 EOM was $2.54 \text{ dB}\cdot\text{cm}^{-1}\cdot\text{GHz}^{-1/2}$, with a good fit to the square-root form over the entire range, whereas that

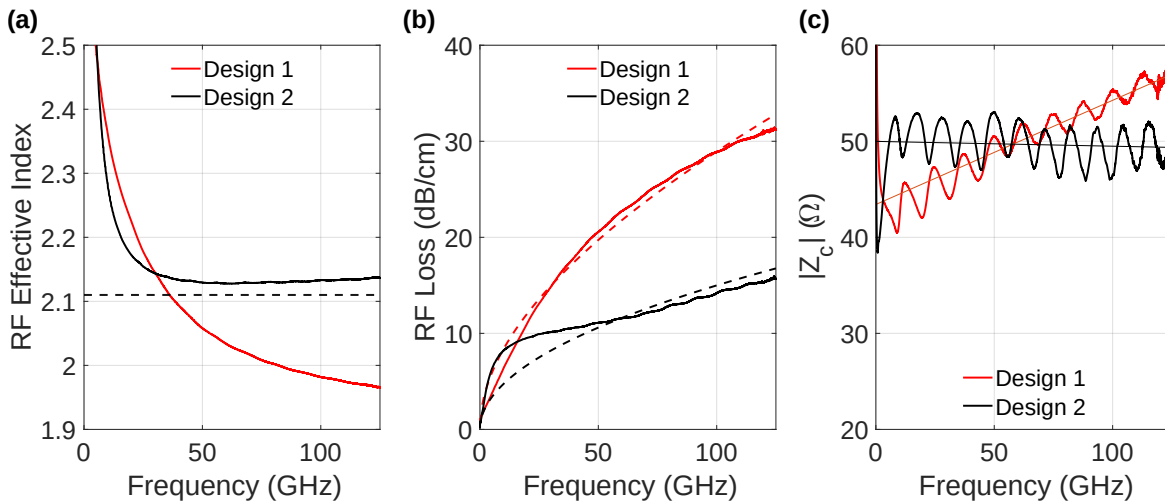


Fig. 7. The extracted RF characteristics of the Design 1 (red) and Design 2 (black) EOM devices from the measured S-parameters. (a), RF effective index. The dashed line shows the target optical group index (2.11). (b), RF propagation loss, where the dashed lines are a fit following the skin-effect model, as described in the text. (c), RF driving impedance, with a linear fit shown by the thin lines.

of the Design 2 EOM was $1.50 \text{ dB} \cdot \text{cm}^{-1} \cdot \text{GHz}^{-1/2}$ (mainly describing the trend above 40 GHz, which is responsible for limiting the EOR 3-dB bandwidth). For comparison, in the hybrid TFLN/Si EOMs with Au electrodes and a high-resistivity Si substrate described in Ref. [46], we reported a lower value of $\alpha_{m0} = 0.7 - 0.8 \text{ dB} \cdot \text{cm}^{-1} \cdot \text{GHz}^{-1/2}$. While Au fundamentally offers lower RF loss, the EOR performance of short EOM devices with Au and Al electrodes can be similar [59]. Since Al is a CMOS-compatible metal, unlike Au, we can anticipate further iterative process improvements, better fidelity between the design and the fabricated structure, and ease of scale-up to larger and more complex designs using foundry processing with Al electrodes and optimized substrate removal.

V. DISCUSSION

In this section, we include a broader discussion of hybrid modulators as part of a PIC, followed by a comparison with earlier results on SiN/TFLN modulators. Various types of hybrid SiN/TFLN modulators are listed in Table I. Missing entries in this table were not reported or could not be inferred from the paper.

These EOM devices were designed primarily for integration as part of a larger PIC, and not as stand-alone fiber-pigtailed devices. Previously, the fiber-to-waveguide coupling loss for the SiN1 layer waveguides in the LIGENTEC process has been measured to be about 1 dB/facet using lensed fibers with a mode field diameter of $2.5 \mu\text{m}$. These values were measured on chips without bonded TFLN. Even lower numbers of about 0.7 dB/facet at 1550 nm were quoted in some references [60]. This reference design for the edge couplers was not changed for these EOM devices and there is no LN above the oxide layer at the edge interfaces. Therefore, the potential fiber-to-fiber insertion loss can be estimated as about 1.5–2 dB higher than the on-chip IL stated earlier. However, a comprehensive study of total IL on the bonded chips is a subject of future study.

The hybrid optical mode shown in Fig. 3(g) is suitable for the straight phase-shift sections but does not support tight bends, for which the mode shown in Fig. 3(c) should be used. This SiN1 layer mode can be used, for example, to form microring resonators [61], [62] or folded modulators.

Recently, wafer-scale bonding of TFLN with single-layer LPCVD SiN waveguides was shown, but with relatively thin SiN layers (thickness of 200 nm or less) and lacking a report of EO performance [63]. The thicker SiN layers used here are more challenging to integrate, but have broader utility for low-loss waveguides, dispersion engineering, and nonlinear PIC components [64].

Also, compared to the hybrid LPCVD SiN modulators reported in Ref. [65] ($V_\pi L = 8.8 \text{ V} \cdot \text{cm}$ with $G = 7 \mu\text{m}$ for a single-arm design, from which we can infer $V_\pi L = 4.4 \text{ V} \cdot \text{cm}$ for a push-pull modulator), the devices reported here improve substantially on both efficiency and bandwidth. We show a 3-dB EO bandwidth beyond 110 GHz whereas no high-bandwidth modulation was shown in Ref. [65]. We believe the $V_\pi L$ value can be reduced by redesign such as reducing the signal-electrode gap without making major changes to the platform.

To overcome the inherent limitations of a single-layer SiN design, dual-layer hybrid SiN/TFLN modulators were developed using PECVD SiN waveguides and a die-bonding approach [16]. Here, we used LPCVD SiN waveguides and wafer-scale bonding, for more scalability in the fabrication process as well as improved performance. For a similar push-pull MZI modulator of length 5 mm, Ref. [16] demonstrated about 31 GHz 3-dB EO bandwidth and $V_\pi L = 6.7 \text{ V} \cdot \text{cm}$. Our results show a large improvement in bandwidth from 31 GHz to greater than 110 GHz for short modulators, and a reduction (i.e., improvement) in $V_\pi L$ from 6.7 to 3.8 V·cm. The insertion loss of about 13 dB in Ref. [16] near 1550 nm is greatly reduced to 4 dB here.

Finally, we emphasize that this work incorporates high-performance EOMs with a multi-layer photonics platform in

a modular way. Designs, materials, and fabrication processes were selected to not adversely impact previously-established components, and to lower overall risk. While exhaustive reliability and repeatability data is not yet available, LIGENTEC has run several wafer lots already with favorable outcomes. The bonding-based film transfer technology and hybrid-mode design approach are broadly useful, since other portions of the same wafer have passive waveguide components, such as the waveguide-to-fiber couplers, inter-layer mode transitions, and the 2×2 couplers, which will not change significantly even if other EO materials replace LN in the future. We can anticipate iterative improvements of performance and scalability to larger and more powerful PICs. These advances can benefit a wide range of users across many application areas, including fiber and free-space communications; analog waveform acquisition, generation, and processing; sensing; displays and beamforming; and quantum photonics technologies.

VI. CONCLUSION

In conclusion, we have described a platform for integrating TFLN EO modulators with low-loss LPCVD SiN waveguides made in a foundry process starting with standard (low-resistivity) silicon wafers. The optical waveguide routing, splitting, and tapering was implemented using the SiN waveguides and no waveguide patterning was necessary on the LN layer. We have reported the optical and RF loss parameters and described the hybrid-mode design as applied to a representative Mach-Zehnder EO modulator. The fabricated devices offer an attractive combination of superior performance metrics, with extinction ratio greater than 35 dB, a half-wave voltage - length product ($V_{\pi}L$) around 3.8 V.cm, about 4 dB on-chip insertion loss and a high-frequency 3-dB EO modulation bandwidth of 110 GHz for short modulators. These results show that scalable silicon-based wafer processing can be used to seamlessly integrate high-performance TFLN EO devices with a low-loss SiN-based integrated photonics platform.

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TABLE I
HYBRID SiN/TFLN ELECTRO-OPTIC MODULATORS

Citation	SiN Technology	SiN Type	Si Substrate	Electrodes	$V_{\pi}L$ (V.cm)	V_{π} (V)	L (mm)	BW (GHz)	IL (dB)	ER (dB)
[32]	Deposition	PECVD	LN substrate	Gold	3	2.5	12	8	8.4	14
[16]	Bonding (die)	PECVD	High resistivity	Aluminum	6.7	13	5	31	13	-
[37]	Deposition	LPCVD	High resistivity	Gold	3.1	1.3	24	29	1.5	27
[66]	Deposition	Sputtering	Unknown	Gold	2.2	2.8	8	25	-	20
[67]	Deposition	PECVD	Unknown	Gold	3.3	3.0	11	30	10	45
[59]	Bonding (die)	PECVD	High resistivity	Aluminum	2.8	5.6	5	110	16	28
[68]	Bonding (wafer)	PECVD	Unknown	Aluminum	2.6	5.2	5	110	5.8	-
[69]	Transfer printing	Unknown	Standard	Gold	3.0	15	2	52	3.3	-
[70]	PECVD	Bonded (die), etched TFLN	Standard ($< 100 \Omega \cdot \text{cm}$)	Gold	3.0	4.3	7	37	1	33
[26]	Transfer printing	Unknown	Unknown	Gold	3.2	3.2	10	35	4.5	31
This work	Bonding (wafer)	LPCVD	Standard ($5 \Omega \cdot \text{cm}$)	Aluminum	3.8	6.4	6	110	3.6	36
					3.9	4.3	9	70	5.5	35
					3.9	2.6	15	20	6.9	36

Abbreviations. SiN: silicon nitride. LN: lithium niobate. TFLN: thin-film LN. PECVD: plasma-enhanced chemical vapor deposition. LPCVD: low-pressure chemical vapor deposition. BW: 3-dB electro-optic modulation bandwidth. IL: on-chip insertion loss. ER: voltage-driven extinction ratio.

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